ADAPTIQ — ADAPTIVE INTELLIGENCE FOR PERSONALIZED LEARNING

[123] THULIR

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Grounding: 100% Derived from 2 Verified Course Document(s)

PART A: 2-MARK QUESTIONS & VERIFIED ANSWERS

Concise Definitions, Key Invariants, Operational Differences & Formulas

Total Questions: 10 • Marks per Question: 2 Marks • Total Marks: 20 Marks

INDEX OF COVERED TOPICS & LEARNING UNITS:

1. Infrastructure • 2. Information Security • 3. Human-Computer Interaction • 4. Analytics & Metrics • 5. Workflow Management • 6. System Engineering • 7. Project Management • 8. Networking • 9. Embedded Systems

QUESTION 1 Topic: Infrastructure **[2 Marks]**

Q1. Define the Cloud Deployment Topology in Project Thulir.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 42) — "Cloud deployment topology illustrating container clusters, load balancers, and regional failovers...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The cloud deployment topology outlines the structured arrangement of container clusters, load balancers, and regional failovers. It guarantees high availability and robust scalability across multiple operational regions.

QUESTION 2 Topic: Information Security **[2 Marks]**

Q2. What is the function of the Access Control Matrix in information security?

Difficulty: EASY | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 34) — "Access control matrix detailing role hierarchies, permissions, and data encryption standards...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The access control matrix details role hierarchies, permission assignments, and data encryption standards. It enforces strict compliance and security measures to protect sensitive user submissions.

QUESTION 3 Topic: Human-Computer Interaction **[2 Marks]**

Q3. Describe the core UI wireframe components for user engagement.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 25) — "UI wireframes highlighting mentor-mentee matching dashboards and accessibility features...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

UI wireframes highlight mentor-mentee matching dashboards and built-in accessibility features. These components are designed to maximize rural user participation and low-latency interaction.

QUESTION 4 Topic: Analytics & Metrics **[2 Marks]**

Q4. Identify the primary elements displayed in the analytics dashboard mockup.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 12) — "Dashboard mockup displaying key performance indicators, impact metrics, and feedback aggregation charts...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The dashboard mockup displays key performance indicators (KPIs), impact metrics, and feedback aggregation charts. These elements provide a quantitative framework for assessing project outcomes.

QUESTION 5 Topic: Workflow Management **[2 Marks]**

Q5. Explain the role of the stage-gate evaluation flowchart in workflow management.

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 18) — "Stage-gate evaluation flowchart mapping ideation filtering down to prototyping approval...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The stage-gate evaluation flowchart maps out ideation filtering down to prototyping approval. It acts as an end-to-end processing pipeline that enforces rigorous state validation rules at each transition stage.

QUESTION 6 Topic: System Engineering **[2 Marks]**

Q6. What does the high-level system design of Project Thulir showcase?

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 5) — "High-level system design showcasing core modular boundaries and stakeholder interfaces for Project Thulir...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

It showcases core modular boundaries and stakeholder interfaces for Project Thulir. This establishes the overall system invariant through structured data flow and module separation.

QUESTION 7 Topic: Project Management **[2 Marks]**

Q7. What logistical considerations are outlined in the deployment roadmap?

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 42) — "Deployment roadmap and logistical considerations for scaling across multiple agrarian districts...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The deployment roadmap covers infrastructure scaling blueprints and logistical steps necessary for scaling operations across multiple agrarian districts.

QUESTION 8 Topic: Human-Computer Interaction **[2 Marks]**

Q8. How do design frameworks enhance rural user accessibility in the system?

Difficulty: EASY | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 25) — "Design frameworks emphasizing voice-first navigation for rural user accessibility...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Design frameworks emphasize voice-first navigation paradigms to overcome literacy barriers and maximize participation among rural innovators and mentors.

QUESTION 9 Topic: Networking **[2 Marks]**

Q9. State the telemetry transmission protocols utilized for networking in constrained environments.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 30) — "Telemetry transmission protocols utilizing MQTT over constrained cellular links...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The system utilizes MQTT protocols operating over constrained cellular links for efficient telemetry transmission.

QUESTION 10 Topic: Embedded Systems **[2 Marks]**

Q10. Describe the function of automated feedback loops in embedded systems.

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 12) — "Automated feedback loops governing water valve controllers based on real-time soil moisture metrics...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Automated feedback loops govern water valve controllers based on real-time soil moisture metrics, ensuring precise resource utilization and responsive agrarian management.